Equivariant Spaces

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Abstract

Contents

1 Topological Groups

1.1 Topological Groups and the Coset Space

Definition 1.1.1 (Topological Groups) Let G be a group. We say that G is a topological group if G is also a topological space and that the following are true.
- The multiplication map $\cdot : G \times G \to G$ defined by $(g, h) \mapsto gh$ is continuous.
- The inverse map $(-)^{-1} : G \to G$ defined by $g \mapsto g^{-1}$ is continuous.

Notice that every group can be given the discrete topology, and so every group is trivially a topological group. But of course there is no guarantee that anything interesting theorems will occur in this case. We call these topological groups discrete.

Definition 1.1.2 (Discrete Group) Let G be a topological group. We say that G is a discrete group if it has the discrete topology.

Proposition 1.1.3
Let G be a topological group. Let $H \leq G$ be a subgroup. Then H is a topological group.

Definition 1.1.4 (Continuous Homomorphisms)
Let G, H be topological groups. Let $\varphi : G \to H$ be a function. We say that φ is a continuous homomorphism if the following are true.
- φ is a group homomorphism.
- φ is continuous.

Definition 1.1.5 (Topological Group Isomorphism)
Let G, H be topological groups. Let $\varphi : G \to H$ be a continuous homomorphism. We say that φ is a topological group isomorphism if there exists a continuous homomorphism $\psi : H \to G$ such that

$$\psi \circ \varphi = \text{id}_G \quad \text{and} \quad \varphi \circ \psi = \text{id}_H$$

1.2 Equivariant Spaces

Definition 1.2.1 (Continuous Group Actions)
Let G be a topological group. Let X be a space. A continuous group action of G on X is a function $\cdot : G \times X \to X$ such that it is a group action and is continuous. In this case, we say that X is a G-space.

Frequently a continuous group action is also called a (topological) transformation group, for example in Milnor's Topology of Fiber Bundles or Introduction to Compact Topological Groups.

Definition 1.2.2 (Group of Homeomorphisms)
Let X be a space. Define the group of homeomorphisms of X to be

$$\text{Homeo}(X) = \{ f : X \to X \mid X \text{ is a homeomorphism} \}$$

Proposition 1.2.3
Let G be a topological group. Let X be a G-space. Then the map $G \to \text{Homeo}(X)$ defined by $g \mapsto (x \mapsto g \cdot x)$ is a group homomorphism.

Let $\varphi : G \to \text{Homeo}(X)$ be a group homomorphism. Then the induced group action $g \mapsto (x \mapsto g \cdot x)$ may not necessarily be continuous. Hence the set of continuous group actions is not quite the set of group homomorphisms from G to $\text{Homeo}(X)$. This is the same as saying not every group action of G on X is continuous.

Definition 1.2.4 (G-Equivariant Maps)
Let G be a topological group. Let X, Y be G-spaces. Let $f : X \to Y$ be a map. We say that f is G-equivariant if we have
$$f(g \cdot x) = g \cdot f(x)$$
for all $x \in X$ and $g \in G$.

Example 1.2.5
Let G, H be topological groups. Let $f : G \to H$ be a function. Give H the structure of a G-space by the action $g \cdot h = f(g)h$. Then f is a G-equivariant map if and only if f is a continuous homomorphism.

Thus the theory of equivariant spaces subsumes the theory of topological groups.

Definition 1.2.6 (Equivariant Isomorphisms)
Let G be a topological group. Let X, Y be G-spaces. Let $f : X \to Y$ be a G-equivariant map. We say that f is an equivariant isomorphism if there exists a G-equivariant map $h : Y \to X$ such that
$$h \circ f = \mathrm{id}_X \quad \text{and } f \circ h = \mathrm{id}_Y$$

Definition 1.2.7 (Homogeneous Space)
Let G be a topoglogical group. Let X be a G-space. We say that X is a homogeneous G-space if G acts transitively on X.

Proposition 1.2.8
Let G be a topological group. Then G is a homogeneous G-space with the action of left multiplication.

1.3 The Coset Space and The Isomorphism Theorems

Definition 1.3.1 (The Coset Space)
Let G be a topological group. Let $H \leq G$ be a subgroup. Define the coset space of H in G to be the quotient space
$$G/H = \{gH \mid g \in G\}$$
whose equivalent relation is given by $g_1 \sim g_2$ if $g_1 g_2^{-1} \in H$

Being a quotient space, the coset space comes equipped with the universal property: Let $p : G \to G/H$ be the quotient map. For any continuous map $f : G \to X$ to a space X such that $f(h_1) = f(h_2)$ for $h_1, h_2 \in H$, there exists a unique map $\widetilde{f} : G/H \to X$ such that the following diagram commutes:
$$\begin{tikzcd} G \arrow[r, "p"] \arrow[dr, "f"'] & G/H \arrow[d, dashed, "\exists! \widetilde{f}"] \\ & X \end{tikzcd}$$

Proposition 1.3.2
Let G be a topological group. Let $H \leq G$ be a subgroup. Consider the action of G on G/H by left multiplication. Then G/H is a homogeneous G-space.

Proposition 1.3.3
Let G be a topological group. Let $H, K \leq G$ be subgroups. Then the following are true.
- There exists a G-equivariant map $G/H \to G/K$ if and only if H and K are conjugate.
- Let $\varphi : G/H \to G/K$ be a G-equivariant map. Then there exists $g \in G$ such that $gHg^{-1} = K$ and $\varphi(aH) = gaH$.
- Let $\varphi_g : G/H \to G/K$ be the map given by $\varphi_g(aH) = gaH$ for $g \in G$ such that $gHg^{-1} = K$.

Then $\varphi_g = \varphi_h$ if and only if $gh^{-1} \in K$.

Proposition 1.3.4
Let G be a topological group. Let $H \trianglelefteq G$ be a normal subgroup. Then G/H is a topological group.

If H is a normal subgroup, then the universal property of quotient spaces and the universal property of quotient groups give the following universal property: For any topological group K and any continuous homomorphism $f : G \to K$ such that $f(h) = 1_K$ for all $h \in H$, there exists a unique continuous homomorphism $\widetilde{f} : G/H \to K$ such that the following diagram commutes:

$$\begin{tikzcd} G \arrow[r, "p"] \arrow[dr, "f"'] & G/H \arrow[d, dashed, "\exists ! \widetilde{f}"] \\ & K \end{tikzcd}$$

Let G, H be a topological groups. Let $\varphi : G \to H$ be a continuous homomorphism. By the first isomorphism theorem of groups, there is a group isomorphism $G/\ker(\varphi) \cong \text{im}(\varphi)$ induced by φ. In general, the induced map $G/\ker(\varphi) \to \text{im}(\varphi)$ is only a continuous homomorphism that is a bijection, and hence not a topological group isomorphism.

Theorem 1.3.5 (The First Isomorphism Theorem for Topological Groups)
Let G, H be topological groups. Let $\varphi : G \to H$ be a continuous homomorphism. If G is locally compact, then the induced map

$$\overline{\varphi} : \frac{G}{\ker(\varphi)} \to \text{im}(\varphi)$$

is a topological group isomorphism.

Theorem 1.3.6 (The Second Isomorphism Theorem for Topological Groups)
Let G be a topological group. Let $H, K \trianglelefteq G$ be normal subgroups such that $H \subseteq K$. Then there is a topological group isomorphism

$$\frac{G/H}{K/H} \cong G/K$$

induced by the map $G/H \to G/K$ given by $gH \mapsto gK$.

Proposition 1.3.7
Let G be a topological group. Let $H, K \leq G$ be subgroups of G such that $H \leq K$. Then the following are true.

- The topology of K/H as the quotient space of K and the topology of K/H as a subgroup of G/H agree.
- Every subgroup of G/H is isomorphic as a topological groups as a quotient group of K/H.

Warning: There is no third isomorphism theorem for topological groups. If it were true, it would describe an isomorphism

$$\frac{H}{H \cap K} \cong \frac{H + K}{K}$$

where $H, K \leq G$ are subgroups such that $K \trianglelefteq G$ (we use addition here as the group operation on G). We construct a counter example for this statement. Let $G = \mathbb{R}$, $K = \mathbb{Z}$ and $H = \lambda\mathbb{Z}$ for $\lambda \in \mathbb{R} \setminus \mathbb{Q}$. Then $H \cap K = \{0\}$ so the left hand side is isomorphic to $\mathbb{Z}$ equipped with the discrete topology. If $\mathbb{Z} + \lambda\mathbb{Z}$ is dense in $\mathbb{R}$, then the right handside evaluates to $\frac{\mathbb{Z}+\lambda\mathbb{Z}}{\mathbb{Z}}$ as a dense subspace of $\mathbb{R}/\mathbb{Z} \cong S^1$. But it does not carry the discrete topology since every open set of S^1 intersects $\frac{\mathbb{Z}+\lambda\mathbb{Z}}{\mathbb{Z}}$ at infinitely many points, so that no finite set is open.

So it remains to show that $\mathbb{Z} + \lambda\mathbb{Z}$ is dense in $\mathbb{R}$.

1.4 Fundamental Systems of Neighbourhoods

Definition 1.4.1 (Fundamental Systems of Neighbourhoods)
Let X be a space. Let $x \in X$. A fundamental system of neighbourhoods of x consists of a collection of open neighbourhoods

$$\mathcal{F}_x \subseteq \{U \subseteq X \mid U \text{ is an open neighbourhood of } x\}$$

of x such that for any open neighbouhood $V \subseteq X$ of x, there exists $U \in \mathcal{F}$ such that $U \subseteq V$.

Lemma 1.4.2
Let X be a space. Let $\mathcal{F}_x$ be a fundamental system of neighbourhoods of x for each $x \in X$. Then $\bigcup_{x\in X} \mathcal{F}_x$ is a basis for X.

Let G be a topological group. Let $\mathcal{F}$ be a fundamental system of neighbourhoods of 1_G. Then for any $g \in G$, the set $\mathcal{F}_g = \{gU \mid U \in \mathcal{F}\}$ is a fundamental system of neighbourhoods of g since left multiplication by g is a homeomorphism. This is the first example of how the local topology of g can be understood just by looking locally at 1_G because G is a homogeneous G-space with the action of left multiplication.

Proposition 1.4.3
Let G be a topological group. Let $\mathcal{F}$ be a fundamental system of neighbourhoods at 1_G. Then the following are true.

- FN1: If $U, V \in \mathcal{F}$, then there exists $W \in \mathcal{F}$ such that $W \subseteq U \cap V$.
- FN2: If $g \in U \in \mathcal{F}$, then there exists $V \in \mathcal{F}$ such that $gV \subseteq U$.
- FN3: If $U \in \mathcal{F}$, then there exists $V \in \mathcal{F}$ such that $V^{-1}V \subseteq U$.
- FN4: If $U \in \mathcal{F}$ and $g \in G$, then there exists $V \in \mathcal{F}$ such that $g^{-1}Vg \subseteq U$.

Proposition 1.4.4
Let G be a group. Let $\mathcal{F} \subseteq \mathcal{P}(G)$ be a collection of subsets of G such that for each $U \in \mathcal{F}$, we have $1_G \in \mathcal{F}$ and that the following are satisfied:

- FN1: If $U, V \in \mathcal{F}$, then there exists $W \in \mathcal{F}$ such that $W \subseteq U \cap V$.
- FN2: If $g \in U \in \mathcal{F}$, then there exists $V \in \mathcal{F}$ such that $gV \subseteq U$.
- FN3: If $U \in \mathcal{F}$, then there exists $V \in \mathcal{F}$ such that $V^{-1}V \subseteq U$.
- FN4: If $U \in \mathcal{F}$ and $g \in G$, then there exists $V \in \mathcal{F}$ such that $g^{-1}Vg \subseteq U$.

Then there exists a unique topology on G such that G is a topological group and $\mathcal{F}$ is a fundamental system of neighbourhoods of 1_G.

As described above, the local topology of G can be understood by the properties around 1_G. Therefore the connected component of 1_G plays a special role.

Definition 1.4.5 (The Identity Component)
Let G be a topological group. Define the identity component of G to be

$$G_0 = \bigcup_{1_G \in U \text{ is connected}} U$$

1.5 Topological Properties of Topological Groups

We investigate topological properties of topological groups, and the relation of topological properties between its subgroups and quotient groups. This includes the separation axioms, connectedness and compactness.

We begin with a discussion of Hausdorffness.

Proposition 1.5.1
Let G be a topological group. Let $H \leq G$ be a subgroup. Then the following are true.
- H and $\overline{H}$ are topological groups.
- If H is normal, then $\overline{H}$ is normal.
- $N_G(H)$ and $C_G(H)$ are closed subgroups of G.

Proposition 1.5.2
Let G be a topological group. Let $H \leq G$ be a subgroup. Then the following are true.
- If H is open, then H is closed.
- If H is closed and $[G : H] < \infty$, then H is open.
- If $U \subseteq G$ is open and $1_G \in U \subseteq H$, then H is open.

Let X be a space. Recall that X is T_1 if $\{x\}$ is a closed subset for all $x \in X$.

Proposition 1.5.3
Let G be a topological group. Then the following are equivalent.
- G is Hausdorff.
- For any topological group H and any continuous homomorphism $\varphi : H \to G$, we have that $\ker(\varphi)$ is a closed subgroup of H.
- $\{1_G\}$ is a closed subgroup of G.
- G is a T_1 space.
- For any fundamental system of neighbourhoods $\mathcal{F}$ of 1_G, we have
$$\bigcap_{U\in\mathcal{F}} U = \{1_G\}$$
- We have
$$\bigcap_{1_G\in U \text{ is open}} U = \{1_G\}$$

Let X be a space. Recall that X is T_0 if for any $x, y \in X$, there exists an open set $U \subseteq X$ such that $x \in U$ or $y \in U$ and not both.

Proposition 1.5.4
Let G be a topological group. Let G be a T_0 space. Then G is Hausdorff.

Proposition 1.5.5
Let G be a topological group. Let $H \leq G$ be a subgroup. Then the following are true.
- G/H is Hausdorff if and only if H is closed in G.
- If H and G/H are Hausdorff, then G is Hausdorff.
- G/H is discrete if and only if H is open in G.

Let X be a space. Recall that X is regular if one of the following equivalent conditions are satisfied:

- For all $x \in X$, and all open set $U \subseteq X$ such that $x \in U$, there exists an open set $V \subseteq U$ containing x such that $\overline{V} \subseteq U$.
- For all $x \in X$ and any closed set C such that $x \notin C$, there exists disjoint open subsets $U, V \subseteq X$ such that $x \in U$ and $C \subset V$.

Proposition 1.5.6
Let G be a topological group. Then the following are true.
- G is regular.
- If $H \leq G$ is a subgroup, then G/H is regular.

Proposition 1.5.7
Let G be a topological group. Let $H \leq G$ be a subgroup. Then the following are true.
- If G is connected, then G/H is connected.
- If H and G/H are connected, then G is connected.
- If H and G/H are totally disconnected, then G is totally disconnected.

Lemma 1.5.8
Let G be a topological group. If G is connected, then G has no proper open subgroups.

Proposition 1.5.9
Let G be a topological group. Then the following are true regarding the identity component.
- G_0 is a closed normal subgroup of G.
- G/G_0 is totally disconnected and Hausdorff.
- The connected components of G are the cosets of G_0.
- Every open subgroup of G contains G_0.
- G_0 is compact.
- We have
$$G_0 = \bigcap_{\substack{H \leq G \text{ is open} \\ \text{and normal}}} H$$

Proposition 1.5.10
Let G be a topological group. Let $H \leq G$ be a subgroup. Then the following are true.
- If G is compact, then $[G : H] < \infty$.
- If G is compact and H is closed, then H is compact.
- If H and G/H are compact, then G is compact.

Proposition 1.5.11
Let G be a topological group. Let G_0 be the connected component of G containing 1_G. Then G is locally compact if and only if there exist $1_G \subseteq U \subseteq G$ such that U is compact.

Proposition 1.5.12
Let G be a topological group. Let $H \leq G$ be a subgroup. Then the following are true.
- If G is locally compact, then G/H is locally compact.
- If H and G/H are locally compact, then G is locally compact.

2 Equivariant Topology

2.1 More on Homogeneous Spaces

Proposition 2.1.1
Let G be a locally compact topological group. Let X be a G-space. Let $x_0 \in X$. Suppose that $\text{Stab}_G(x_0)$ is a closed subgroup. Then there is an isomorphism of G-spaces

$$\frac{G}{\text{Stab}_G(x_0)} \cong \text{Orb}_G(x_0)$$

induced by the map $p : G \to \text{Orb}_G(x_0)$ given by $g \mapsto g \cdot x_0$.

Proposition 2.1.2
Let G be a locally compact topological group. Let X be a Hausdorff G-space. Let $x_0 \in X$. Suppose that $\text{Stab}_G(x_0)$ is a closed subgroup. Then the following are true.

- If X is a homogeneous G-space, then there is an isomorphism of G-spaces
$$\frac{G}{\text{Stab}_G(x_0)} \cong X$$
induced by the map $g \mapsto g \cdot x_0$.
- If X is a homogeneous G-space and the action of G on X is free, then there is an isomorphism of G-spaces
$$G \cong X$$
given by the map $g \mapsto g \cdot x_0$.

Proposition 2.1.3 Let G be a topological group and let X be a homogenous G-space. Then the following are true.

- For any $\varphi \in \text{Homeo}(X)$, $x \in X$ and $\varphi(x)$ has the same stabilizers
- If $x, y \in X$ has the same stabilizers, then there exists $\varphi \in \text{Homeo}(X)$ such that $\varphi(x) = y$.

Lemma 2.1.4 Let G be a topological group and let X be a homogenous G-space. Let A be a subgroup of $\text{Homeo}(X)$. Then $A = \text{Homeo}(X)$ if and only if for any $x, y \in X$ such that $\text{Stab}_G(x) = \text{Stab}_G(y)$, there exists $\varphi \in A$ such that $\varphi(x) = y$.

Proposition 2.1.5 Let G be a topological group and let X be a homogenous G-space. Then there is a isomorphism of left G-spaces

$$\frac{N(\text{Stab}_G(x_0))}{\text{Stab}_G(x_0)} \cong \text{Homeo}(X)$$

2.2 Orbit Spaces

Definition 2.2.1 (The Orbit Space)
Let G be a topological group. Let X be a G-space. Define the orbit space of X by G to be the quotient space

$$X/G = \{\text{Orb}_G(x) \mid x \in X\}$$

of X whose equivalent relation is given by $x \sim y$ if $gx = y$ for some $g \in G$.

Let G be a topological group. Let $H \leq G$ be a subgroup. We discussed extensively the topology of G/H defined by the quotient map $p : G \to G/H$. As usual, there is viewpoint of group actions. Consider the action of H on G by left multiplication. Then the orbit space is precisely G/H. Thus once again the theory of G-spaces subsumes the theory of topological groups.

Proposition 2.2.2
Let G be a topological group. Let X be a G-space. Suppose that G and X are Hausdorff, and G is compact. Let $p : X \to X/G$ be the quotient map. Then the following are true.
- p is a closed map.
- p is a proper map.
- X is compact if and only if X/G is compact.
- X is locally compact if and only if X/G is locally compact.

2.3 Proper Group Actions

Let G be a group. Let X be a space. We would like to ask for criteria in which X/G is a Hausdorff.

Proposition 2.3.1
Let G be a group. Let X be a space. Let $\varphi : G \to \text{Homeo}(X)$ be a group homomorphism. Then X/G is Hausdorff if and only if for all $[g], [h] \in X/G$ distinct, there exists neighbourhoods $U, V \subseteq X$ of g and h respectively such that $U \cap (k \cdot V) = \emptyset$ for all $k \in G$.

This is the most general necessary and sufficient criterion. However, it is not an easy criterion to check. Therefore we turn to the notion of proper group action.

Definition 2.3.2 (Proper Group Actions)
Let G be a topological group. Let X be a G-space. We say that the action of G on X is proper if the map $G \times X \to X \times X$ given by
$$(g, x) \mapsto (x, g \cdot x)$$
is a proper map.

Lemma 2.3.3
Let G be a compact topological group. Let X be a Hausdorff G-space. Then the action of G on X is proper.

Proposition 2.3.4
Let G be a topological group. Let X be a Hausdorff G-space. Then the action of G on X is proper if and only if for every compact subset $K \subseteq X$, the set
$$\{g \in G \mid (g \cdot K) \cap K = \emptyset\}$$
is compact.

Proposition 2.3.5
Let G be a topological group. Let X be a locally compact Hausdorff G-space. If the action of G on X is proper, then X/G is Hausdorff. (Lee Topological Manifolds 12.24)

Proposition 2.3.6
Let G be a topological group. Let X be a G-space. Suppose that G and X are Hausdorff and X is locally compact. Suppose that the action of G on X is proper. Then the following are true.
- The map $f : G \to X$ given by $g \mapsto g \cdot x$ is proper for any $x \in X$.
- $\text{Stab}_G(x)$ is compact for any $x \in X$.
- The map
$$\frac{G}{\text{Stab}_G(x)} \to \text{Orb}_G(x)$$
induced by f is a homeomorphism for any $x \in X$.
- $\text{Orb}_G(x)$ is closed in X for all $x \in X$.

(Tom Dieck Transformation Groups 3.19)

2.4 Covering Space Actions

Let G be a group (not necessarily topological). Let X be a space. We ask the question of when is the projection map $p : X \to X/G$ a covering space.

Definition 2.4.1 (Covering Space Actions)
Let G be a group. Let X be a space. Let $\varphi : G \to \text{Homeo}(X)$ be a group homomorphism. We say that the action of G on X is a covering space action if for all $x \in X$, there exist a neighbourhood $U \subseteq X$ of x such that

$$U \cap (g \cdot U) \neq \emptyset$$

implies $g = 1_G$.

Lemma 2.4.2
Let G be a group. Let X be a space. Let $\varphi : G \to \text{Homeo}(X)$ be a group homomorphism. Then the action of G on X is a covering space action if and only if for each $x \in X$, there exists a neighbourhood $U \subseteq X$ of x such that

$$(g_1 \cdot U) \cap (g_2 \cdot U) \neq \emptyset$$

implies $g_1 = g_2$.

Proof
True since $(g_1 \cdot U) \cap (g_2 \cdot U) \neq \emptyset$ if and only if $U \cap (g_1^{-1} g_2 \cdot U) \neq \emptyset$. ■

Definition 2.4.3 (Wandering Actions)
Let G be a group. Let X be a space. Let $\varphi : G \to \text{Homeo}(X)$ be a group homomorphism. We say that the action of G on X is a wandering action if for all $x \in X$, there exists a neighbourhood $U \subseteq X$ of x such that $(g \cdot U) \cap U \neq \emptyset$ for finitely many $g \in G$.

Proposition 2.4.4
Let G be a group. Let X be a space. Let $\varphi : G \to \text{Homeo}(X)$ be a group homomorphism.

- If the action of G on X is a covering space action, then the action is free and wandering.
- If the action of G on X is free and wandering, and X is Hausdorff, then the action is a covering space action.

Proof
Suppose that $g \cdot x = x$ for some $x \in X$ and $g \in G$. Since the action is a covering space action, there exists a neighbourhood $U \subseteq X$ of x such that $U \cap (g \cdot U) \neq \emptyset$ implies $g = 1_G$. But indeed, $x = g \cdot x \in g \cdot U$ implies $x \in U \cap (g \cdot U)$. Hence $g = 1_G$. Clearly covering space actions are wandering. ■

Proposition 2.4.5
Let X be a space. Let $p : \widetilde{X} \to X$ be a covering space of X. Then the action of $\text{Deck}(p)$ on $\widetilde{X}$ is a covering space action.

Proof
Let $x \in \widetilde{X}$. Let $\widetilde{U} \subseteq \widetilde{X}$ be an open neighbourhood of $\widetilde{x}$ such that p induce a homeomorphism from $\widetilde{U}$ to $U = p(\widetilde{U})$. Suppose that $\widetilde{U} \cap f(\widetilde{U}) \neq \emptyset$ for some $f \in \text{Deck}(p)$. Then there exists $y_1, y_2 \in \widetilde{X}$ such that $y_1 = f(y_2)$. Since f is a deck transformation, there exists $z \in X$ such that $y_1, y_2 \in p^{-1}(z)$. But $p|_{\widetilde{U}} : \widetilde{U} \to U$ is a homeomorphism. Hence $y_1 = y_2$. By covering space theory, f fixing one point implies that $f = \text{id}_{\widetilde{X}}$. Hence we are done. ■

Proposition 2.4.6
Let G be a group. Let X be a space. Let $\varphi : G \to \text{Homeo}(X)$ be a group homomorphism. Then the quotient map $X \to X/G$ is a covering space if and only if the action of G on X is a covering

space action.

Proof
Suppose that $X \to X/G$ is a covering space. Then for each $g \in G$, we have $g \cdot - : X \to X$ is clearly a deck transformation. By definition, we have $g_1 \cdot - = g_2 \cdot -$ if and only if $g_1 = g_2$. Hence G is a subgroup of $\text{Deck}(p)$. Hence G is also a covering space action.

Let $x \in X$. Let $U \subseteq X$ be the open neighbourhood of x given by the covering space action. By definition of the quotient map, for each $g \in G$, the quotient map sends $g \cdot U$ homeomorphically to $p(U)$. Hence we are done.

■

Proposition 2.4.7
Let G be a group. Let X be a space. Let $\varphi : G \to \text{Homeo}(X)$ be a group homomorphism. Suppose that the action of G on X is a covering space action. Then the following are true.

- The quotient map $p : X \to X/G$ is a regular covering space.
- If X is path connected, then there is a group isomorphism $G \cong \text{Deck}(p)$.
- If X is path connected and locally path connected, then there is a group isomorphism
$$G \cong \frac{\pi_1(X/G, x_0)}{p_*(\pi_1(X, \widetilde{x_0}))}$$
for any $x_0 \in X$ and $\widetilde{x_0} \in p^{-1}(x_0)$.

Proof
We have seen that p is a covering space. It is regular since $g_2^{-1}g_1$ sends $g_1 \cdot U$ to $g_2 \cdot U$.

For each $g \in G$, the map $g \cdot - : X \to X$ is a deck transformation, and $g_1 \cdot - = g_2 \cdot -$ if and only if $g_1 = g_2$. Hence G is a subgroup of $\text{Deck}(p)$. If X is path connected, let $f \in \text{Deck}(p)$. Then for any $x \in X$, we have $p(x) = p(f(x))$ by definition. Hence $[x] = [f(x)] \in X/G$. Thus there exists $g \in G$ such that $g \cdot x = f(x)$. Since X is path connected, by the unique lifting property, we have $g \cdot - = f$. Hence $G \cong \text{Deck}(p)$.

The last part follows by crl4.2.4 in Algebraic Topology 1. ■

Lemma 2.4.8
Let X be a space. Let $p : \widetilde{X} \to X$ be a regular covering space. Then p induces an homeomorphism
$$\frac{\widetilde{X}}{\text{Deck}(p)} \cong X$$

Proof
Since p is a covering map, we may invoke the universal property of the quotient space $\widetilde{X}/\text{Deck}(p)$ to obtain a map to X. On the other hand, since p is surjective and regular, the function $X \to \widetilde{X}/\text{Deck}(p)$ given by $x \mapsto [p^{-1}(x)]$ is well defined. Since p is sends open sets to open sets, the function is continuous and clearly the desired inverse. ■

Thus the quotient map $X \to X/G$ is a covering space when the action of G on X is a covering space action. But under what conditions is a group action a covering space action? There are a few criteria (We already saw two: free + X Hausdorff + wandering action implies covering space action, covering spaces $X \to X/G$ conversely give covering space actions).

Proposition 2.4.9
Let G be a discrete group. Let X be a locally compact Hausdorff G-space. Suppose that the action of G on X is free. Suppose that one of the following are true.

- For all $K \subseteq X$ compact subsets, the set
$$\{g \in G \mid (g \cdot K) \cap K = \emptyset\}$$
is finite.
- The action of G on X is proper.

Then the following are true.

- The action of G on X is a covering space action.
- X/G is locally compact and Hausdorff.
- $X \to X/G$ is a regular covering space.

Under mild assumptions, proper group actions are also covering space actions.

Proposition 2.4.10
Let G be a group. Let X be a space. Let $\varphi : G \to \text{Homeo}(X)$ be a group homomorphism. Suppose that X/G is Hausdorff and the action of G on X is a covering space action. Then the action of G as a discrete group on X is a proper group action.

Corollary 2.4.11
Let G be a discrete group. Let X be a locally compact and Hausdorff G-space. Then the following are equivalent.

- X/G is Hausdorff and the action of G on X is a covering space action.
- The action of G on X is proper and free.

Proof
Immediate from the above. ■

3 The Covering Space Correspondence

It is easy to see that given a space X, the set $\text{Aut}(X)$ of automorphisms of X form a group. It is often that one studies also the the maps in and out of the object one is studying. In our case, the interesting maps for the covering spaces would be deck transformations.

3.1 Deck Groups

Definition 3.1.1 (Deck Transformations) Let X be a space and $p : \tilde{X} \to X$ a covering space. A deck transformation of p is a homeomorphism $\tau : \tilde{X} \to X$ such that $p \circ \tau = p$. The set of all deck transformations of a cover is called $\text{Deck}(p)$.

The condition that $p \circ \tau = p$ restricts the set of maps that we are considering in $\text{Aut}(\tilde{X})$. In particular, it is a group in its own right, so that $\text{Deck}(p)$ is a subgroup of $\text{Aut}(\tilde{X})$.

Proposition 3.1.2 Let $p : \tilde{X} \to X$ be a covering. Then $\text{Deck}(p)$ is a group.

Proof Let $\tau, \phi \in \text{Deck}(p)$. Then $p\circ(\tau\circ\phi) = (p\circ\tau)\circ\phi = p\circ\phi = p$ which means that the operation is closed. Composition of functions is clearly associativity. The identity map $\text{id} : \tilde{X} \to \tilde{X}$ satisfies the definition of a deck transformation. Finally, since τ is a homeomorphism, it has an inverse, τ^{-1}. Then by pre-composing with τ^{-1}, we get that $p\circ\tau = p$ implies $p = p\circ\tau^{-1}$ thus $\tau^{-1} \in \text{Deck}(p)$. ■

The condition that $p \circ \tau = p$ also restricts what deck transformations can do. As a result, deck transformations are a lot more restrictive compared to simply automophisms. In particular, one can see that for $x \in X$, a deck transformation $\tau : \tilde{X} \to \tilde{X}$ necessarily permutes elements of the fiber $p^{-1}(x_0)$. Moreover, we have the following proposition.

Proposition 3.1.3 Let X be a space and $p : \tilde{X} \to X$ a path connected covering space. Then $\tau \in \text{Deck}(p)$ is the identity if and only if τ has at least one fixed point.

Proof If τ is the identity then τ fixes all points. Now suppose that τ fixes $\tilde{x}_0 \in \tilde{X}$. Let $\tilde{x} \in \tilde{X}$ be an arbitrary point. Since $\tilde{X}$ is path connected, there exists a path $f : I \to \tilde{X}$ from $\tilde{x}_0$ to $\tilde{x}$. Similarly, there exists a path $g : I \to \tilde{X}$ from $\tau(\tilde{x}_0) = \tilde{x}_0$ to $\tau(\tilde{x})$. Since $p \circ \tau = p$, we have that $p \circ f = p \circ g$. By uniqueness of path lifting, f and g both start at $\tilde{x}_0 = \tau(\tilde{x}_0)$ so they also both end at the same point. Thus $\tau(\tilde{x}) = \tilde{x}$ and so we are done. ■

Similar to the fundamental group, the deck group also has a right action on the fibers of a covering space.

Proposition 3.1.4 Let X be a space and $x_0 \in X$. Let $p : (\tilde{X}, \tilde{x}_0) \to (X, x_0)$ be a covering space. Then the map $\cdot : \text{Deck}(p) \times p^{-1}(x_0) \to p^{-1}(x_0)$ defined by

$$\tau \cdot \tilde{x}_0 = \tau(\tilde{x}_0)$$

is a group action on $p^{-1}(x_0)$.

Proposition 3.1.5 Let X be a space and $x_0 \in X$. Let $p : (\tilde{X}, \tilde{x}_0) \to (X, x_0)$ be a covering space. Then the action of $\text{Deck}(p)$ on $p^{-1}(x_0)$ is free.

Proof If $\tau \in \text{Deck}(p)$ is such that $\tau(\tilde{x}) = \tilde{x}$ for some $\tilde{x} \in \tilde{X}$, then by 3.1.3, $\tau = \text{id}$. Thus the group action is free. ■

3.2 Regular Covering Spaces

When there is only one element in the conjugacy class of a subgroup, that group is then a normal subgroup. We will now consider covering spaces such that their induced subgroup of the fundamental group is normal. This classification will also fit in well with our theory of Galois correspondence. We will also see how the deck group is related to normality of the subgroups.

Definition 3.2.1 (Regular Covering Spaces) Let X be a space and $x_0 \in X$. We say that a covering space $p : (\tilde{X}, \tilde{x}_0) \to (X, x_0)$ is regular if $\text{Deck}(p)$ acts transitively on $p^{-1}(x_0)$.

Some authors such as Hatcher calls regular covering spaces as normal covering spaces. The reason is given by the following proposition.

Proposition 3.2.2 Let $p : (\tilde{X}, \tilde{x}_0) \to (X, x_0)$ with $\tilde{X}$ path connected. Then p is a regular covering space if and only if $p_*(\pi_1(\tilde{X}, \tilde{x}_0))$ is a normal subgroup of $\pi_1(X, x_0)$.

Proof Suppose that $\text{Deck}(p)$ is transitive on $p^{-1}(x_0)$. Let $[\gamma] \in \pi_1(X, x_0)$. By by the path lifting property of covering spaces, γ lifts to $\tilde{\gamma}$ where $\tilde{x}_1 = \tilde{\gamma}(1) \in p^{-1}(x_0)$. Since $\text{Deck}(p)$ is transitive, there exists $\tau : (\tilde{X}, \tilde{x}_0) \to (\tilde{X}, \tilde{x}_1)$. By the lifting criterion,

$$\begin{tikzcd} & (\tilde{X}, \tilde{x}_1) \arrow[d, "p"] \\ (\tilde{X}, \tilde{x}_0) \arrow[ur, "\tau"] \arrow[r, "p"'] & (X, x_0) \end{tikzcd}$$

we have that $p_*(\pi_1(\tilde{X}, \tilde{x}_0)) \subseteq p_*(\pi_1(\tilde{X}, \tilde{x}_1))$. Since τ is a homeomorphism, τ^{-1} and the lifting criterion gives

$$p_*(\pi_1(\tilde{X}, \tilde{x}_0)) = p_*(\pi_1(\tilde{X}, \tilde{x}_1))$$

By **??**, we have that

$$[\gamma]p_*(\pi_1(\tilde{X}, \tilde{x}_0))[\overline{\gamma}] = p_*(\pi_1(\tilde{X}, \tilde{x}_1)) = p_*(\pi_1(\tilde{X}, \tilde{x}_0))$$

Since this is true for any $[\gamma] \in \pi_1(X, x_0)$, we conclude that $p_*(\pi_1(\tilde{X}, \tilde{x}_0))$ is a normal subgroup.

Now suppose that $p_*(\pi_1(\tilde{X}, \tilde{x}_0))$ is a normal subgroup. Choose $\tilde{x}_1 \in p^{-1}(x_0)$. Then for any $[\gamma] \in \pi_1(X, x_0)$ such that it lifts to $\tilde{\gamma} : I \to \tilde{X}$ with $\tilde{\gamma}(1) = \tilde{x}_1$, by **??** we have that

$$p_*(\pi_1(\tilde{X}, \tilde{x}_0)) = [\gamma]p_*(\pi_1(\tilde{X}, \tilde{x}_0))[\overline{\gamma}] = p_*(\pi_1(\tilde{X}, \tilde{x}_1))$$

This satisfies the lifting criterion:

$$\begin{tikzcd} & (\tilde{X}, \tilde{x}_1) \arrow[d, "p"] \\ (\tilde{X}, \tilde{x}_0) \arrow[ur, dashed, "\exists!\tau"] \arrow[r, "p"'] & (X, x_0) \end{tikzcd}$$

So there exists a map $\tau : (\tilde{X}, \tilde{x}_0) \to (\tilde{X}, \tilde{x}_1)$. By inverting the orientation of γ, we can invert the results of the lifting criterion so that τ has an inverse. Thus τ is a deck transformation. Since this is true for any $\tilde{x}_1 \in p^{-1}(x_0)$, we conclude. ■

We can now improve on the bijection $\deg(p) = [\pi_1(X, x_0) : p_*(\pi_1(\tilde{X}, \tilde{x}_0))]$, albeit an indirect one. The consequence of the theorem for when p is a regular covering will show that this theorem is indeed a generalization.

Theorem 3.2.3 Let (X, x_0) be a path connected and locally path connected space. Let $p : (\tilde{X}, \tilde{x}_0) \to (X, x_0)$ be a path connected covering space of X. Then we have that

$$\text{Deck}(p) \cong \frac{N(p_*(\pi_1(\tilde{X}, \tilde{x}_0)))}{p_*(\pi_1(\tilde{X}, \tilde{x}_0))}$$

for any $\tilde{x}_0 \in p^{-1}(x_0)$, where N is the normalizer.

As an immediate result, the deck group of a regular covering space $p : \tilde{X} \to X$ is isomorphic to the quotient group with the induced subgroup of p_* since the normalizer of the image subgroup is all of $\pi_1(X, x_0)$ in this case.

Corollary 3.2.4 Let (X, x_0) be a path connected and locally path connected space. Let $p : (\tilde{X}, \tilde{x}_0) \to (X, x_0)$ be a regular covering space of X. Then

$$\text{Deck}(p) \cong \frac{\pi_1(X, x_0)}{p_*\left(\pi_1(\tilde{X}, \tilde{x}_0)\right)}$$

Proof Since $p_*(\pi_1(\tilde{X}, \tilde{x}_0))$ is a normal subgroup in this case, the normalizer is all of $\pi_1(X, x_0)$ and so 3.2.3 implies the result.

But we can also give an independent proof. Define a homomorphism $\phi : \pi_1(X, x_0) \to \text{Deck}(p)$ defined by $[\gamma] \mapsto \tau$ where τ is the unique deck transformation sending $\tilde{\gamma}(0)$ to $\tilde{\gamma}(1)$ (This is possible since $\tilde{X}$ is a regular cover). This is a homomorphism since for $[f]$ and $[g]$ in $\pi_1(X, x_0)$ that lifts to $\tilde{f}$ sending $\tilde{x}_0$ to $\tilde{x}_1$ and $\tilde{g}$ sending $\tilde{x}_1$ to $\tilde{x}_2$, $\phi([f] \cdot [g])$ the unique deck transformation sending $\tilde{x}_0$ to $\tilde{x}_2$. It is clear that $\phi([f]) \circ \phi([g])$ has the same effect.

Notice that this map is surjective. Indeed, given a deck transformation $\tau : (\tilde{X}_0, \tilde{x}_0) \to (\tilde{X}, \tilde{x}_1)$, since $\tilde{X}$ is path connected, there exists a path $\tilde{\gamma} : I \to \tilde{X}$ from $\tilde{x}_0$ to $\tilde{x}_1$. By definition of a deck transformation, $p(\tilde{x}_0) = p(\tilde{x}_1)$ so that $p \circ \tilde{\gamma} = \gamma$ is indeed a loop in X. So $\phi([\gamma]) = \tau$.

We also find the kernel of the map. For $[\gamma] \in \ker(\phi)$, $\phi([\gamma])$ is the identity map of $\tilde{X}$. In particular, $\tilde{\gamma}$ must be a loop in $\tilde{X}$ so that $\tilde{\gamma}$ in $p_*(\pi_1(\tilde{X}, \tilde{x}_0))$. Conversely, given $[\tilde{\theta}] \in p_*(\tilde{X}, \tilde{x}_0)$, $\phi([p \circ \tilde{\theta}])$ is the deck transformation associated to $\tilde{\theta}$ sending $\tilde{\theta}(0)$ to $\tilde{\theta}(1) = \tilde{\theta}(0)$ thus the associated deck transformation is the identity. We conclude that $\ker(\phi) = p_*(\pi_1(\tilde{X}, \tilde{x}_0))$.

The first isomorphism theorem thus implies that

$$\text{Deck}(p) \cong \frac{\pi_1(X, x_0)}{p_*\left(\pi_1(\tilde{X}, \tilde{x}_0)\right)}$$

and so we are done. ∎

Combining **??**, this means that if p is a regular covering, we have that

$$\text{Deck}(p) \cong \frac{\pi_1(X, x_0)}{p_*\left(\pi_1(\tilde{X}, \tilde{x}_0)\right)}$$

implying

$$|\text{Deck}(p)| = \left| \frac{\pi_1(X, x_0)}{p_*\left(\pi_1(\tilde{X}, \tilde{x}_0)\right)} \right| = [\pi_1(X, x_0) : p_*(\pi_1(\tilde{X}, \tilde{x}_0))] = \left|p^{-1}(x_0)\right| = \deg(p)$$

Wow, this has gotten messy.

Corollary 3.2.5 Let (X, x_0) be a path connected and locally path connected space. Let $p : (\tilde{X}, \tilde{x}_0) \to (X, x_0)$ be the universal covering space of X. Then

$$\text{Deck}(p) \cong \pi_1(X, x_0)$$

Proof In addition to being normal, the universal covering space has trivial fundamental group so that 3.2.3 directly implies the result. ■

Combining results from the previous sections, we now have the equality

$$|\pi_1(X, x_0)| = \left|p^{-1}(x_0)\right| = \deg(p) = |\text{Deck}(p)|$$

when $p : (\tilde{X}, \tilde{x}_0) \to (X, x_0)$ is the universal covering space.

3.3 The Correspondence Theorem for Deck Groups

Definition 3.3.1 (Intermediate Covering Spaces)
Let X be a space. Let $p : \widetilde{X} \to X$ and $q : Z \to X$ be covering spaces. We say that q is an intermediate covering space of p if there exists a covering space homomorphism $f : \widetilde{X} \to Z$ such that the following diagram commutes:

$$\begin{tikzcd} \widetilde{X} \arrow[r, dashed, "\exists f"] \arrow[dr, "p"'] & Z \arrow[d, "q"] \\ & X \end{tikzcd}$$

Theorem 3.3.2 (Galois Correspondence for Deck Groups)
Let (X, x_0) be a locally connected space. Let $p : \widetilde{X} \to X$ be a regular covering space. Then there is a one to one inclusion reversing correspondence

$$\{q : Z \to X \mid q \text{ is an intermediate covering space }\} \stackrel{1:1}{\longleftrightarrow} \{H \leq \text{Deck}(p) \mid H \text{ is a subgroup of Deck}(p)\}$$

given as follows.
- An intermediate covering $q : Z \to X$ is sent to $\text{Deck}(q)$.
- An subgroup of $\text{Deck}(p)$ gives the covering space $\widetilde{X}/H$.

When p is the universal covering space, we recover the standard correspondence between the fundamental group and all path connected covering spaces.

Proposition 3.3.3
Let (X, x_0) be a locally connected space. Let $p : \widetilde{X} \to X$ be a regular covering space. The above one to one correspondence induces a bijection

$$\left\{q : Z \to X \;\middle|\; \substack{q \text{ is an intermediate} \\ \text{regular covering space}}\right\} \stackrel{1:1}{\longleftrightarrow} \{H \leq \text{Deck}(p) \mid H \text{ is a normal subgroup of Deck}(p)\}$$

In this case, we have $\text{Deck}(q : \widetilde{X}/H \to X) \cong \frac{\text{Deck}(p)}{H}$.

4 Topology of the General Linear Group

4.1 Subgroups of the Real General Linear Group

Proposition 4.1.1
The group of real $n \times n$ matrices

$$M_n(\mathbb{R}) = \{(a_{ij})_{n\times n} \mid a_{ij} \in \mathbb{R}\}$$

with group structure given by matrix addition can be given the structure of a topological group whose topology is given by the topology of $\mathbb{R}^{n^2}$ and a choice of bijection of sets $\mathbb{R}^{n^2} \cong M_n(\mathbb{R})$.

Proof
Addition is continuous since the formula for each term in the output is a combination of sums and products of terms of the input matrices. ■

Proposition 4.1.2 Let $n \in \mathbb{N} \setminus \{0\}$. Then $GL(n, \mathbb{R})$ is an open subset of $M_n(\mathbb{R})$.

Beware that $GL(n, \mathbb{R}), O(n, \mathbb{R}), SL(n, \mathbb{R}), SO(n, \mathbb{R})$ are not topological groups as subsets of $M_n(\mathbb{R})$ because they are not subgroups of $M_n(\mathbb{R})$ in the first place.

Proposition 4.1.3
Let $n \in \mathbb{N}$. Then $\mathrm{GL}(n, \mathbb{R})$ is a topological group.

Proof
We know that $\mathrm{GL}(n, \mathbb{R})$ is an open subset of $M_n(\mathbb{R}) \cong \mathbb{R}^2$ so we have the induced topology. Multiplication is continuous because multiplication is given by polynomials in each entry, which are continuous. The inverse is continuous since the formula for the inverse consists of determinants of sub matrices, and they are a finite sum and product of terms of the matrices. ■

Proposition 4.1.4
Let $n \in \mathbb{N} \setminus \{0\}$. Then the following are true.
- $O(n, \mathbb{R})$ is a compact topological subgroup of $\mathrm{GL}(n, \mathbb{R})$.
- $\mathrm{SL}(n, \mathbb{R})$ is a topological subgroup of $\mathrm{GL}(n, \mathbb{R})$.
- $\mathrm{SO}(n, \mathbb{R})$ is a compact connected topological subgroup of $\mathrm{GL}(n, \mathbb{R})$.

Moreover, they are closed subsets of $GL(n, \mathbb{R})$.

$GL(n, \mathbb{C}), U(n), SU(n)$

Proposition 4.1.5
Let $n \in \mathbb{N}^+$. Then the action of $O(n+1)$ on S^n induces a homeomorphism

$$\frac{O(n+1)}{O(n)} \cong S^n$$

where we consider $O(n)$ as a subgroup of $O(n+1)$ by the inclusion $A \mapsto \begin{pmatrix} A & 0 \\ 0 & 0 \end{pmatrix}$.

Proof
Clearly S^n is a homogeneous $O(n+1)$-space. By choosing $x_0 = (0, \dots, 0, 1)$, we note that $\mathrm{Stab}_{O(n+1)}((0, \dots, 0, 1)) = O(n)$. So it follows from the above proposition that we have a homeomorphism. ■

Proposition 4.1.6

Let $n \in \mathbb{N}^+$. Then the action of $\text{SO}(n+1)$ on S^n induces a homeomorphism

$$\frac{\text{SO}(n+1)}{\text{SO}(n)} \cong S^n$$

where we consider $\text{SO}(n)$ as a subgroup of $\text{SO}(n+1)$ by the inclusion $A \mapsto \begin{pmatrix} A & 0 \\ 0 & 0 \end{pmatrix}$.

Proof
Same as the above. ■

4.2 Subgroups of the Complex General Linear Group

Example 4.2.1
Let $n \in \mathbb{N}^+$. Then the action of $U(n+1)$ on S^{2n+1} induces a homeomorphism

$$\frac{U(n+1)}{U(n)} \cong S^{2n+1}$$

4.3 Real Stiefel and Grassmannian Space

Definition 4.3.1 (Real Stiefel)
Define the real Stiefel space to be the subspace

$$V_k(\mathbb{R}^n) = \{A \in M_{n\times k}(\mathbb{R}) \mid A \text{ is injective } \}$$

Proposition 4.3.2
There is an $O(n)$-equivariant isomorphism

$$V_k(\mathbb{R}^n) \cong \frac{O(n)}{O(n-k)}$$

Example 4.3.3
We have an isomorphism

$$S^{n-1} \cong V_1(\mathbb{R}^n)$$

Definition 4.3.4 (Real Grassmannian)
Define the real grassmannian to be the quotient space

$$\text{Gr}_k(\mathbb{R}^n) = \frac{V_k(\mathbb{R}^n)}{\sim}$$

where $A \sim B$ if there exists $C \in \text{GL}(k, \mathbb{R})$ such that $A = BC$.

Proposition 4.3.5
There is a natural bijection

$$\text{Gr}_k(\mathbb{R}^n) \overset{1:1}{\longleftrightarrow} \{V \subseteq \mathbb{R}^n \mid V \text{ is a } k\text{-dimensional linear subspace of } \mathbb{R}^n\}$$

Example 4.3.6
We have a homeomorphism

$$\mathbb{RP}^{n-1} \cong \text{Gr}_1(\mathbb{R}^n)$$

given by the above.

Proposition 4.3.7
There is an $O(n)$-equivariant isomorphism

$$\text{Gr}_k(\mathbb{R}^n) \cong \frac{O(n)}{O(k) \times O(n-k)}$$

given by the projection map $V_k(\mathbb{R}^n) \to \text{Gr}_k(\mathbb{R}^n)$.

4.4 Complex Stiefel and Grassmannian Space

Definition 4.4.1 (Complex Stiefel)
Define the complex Stiefel space to be the subspace

$$V_k(\mathbb{C}^n) = \{A \in M_{n\times k}(\mathbb{C}) \mid A \text{ is injective } \}$$

Definition 4.4.2 (Complex Grassmannian)
Define the complex grassmannian to be the quotient space

$$\text{Gr}_k(\mathbb{C}^n) = \frac{V_k(\mathbb{C}^n)}{\sim}$$

where $A \sim B$ if there exists $C \in \text{GL}(k, \mathbb{C})$ such that $A = BC$.

Example 4.4.3
We have a homeomorphism

$$\mathbb{CP}^{n-1} \cong \text{Gr}_1(\mathbb{C})$$

Proposition 4.4.4
There is an equivariant isomorphism

$$V_k(\mathbb{C}^n) \cong \frac{U(n)}{U(n-k)}$$

Proposition 4.4.5
There is an equivariant isomorphism

$$\text{Gr}_k(\mathbb{C}^n) \cong \frac{U(n)}{U(k) \times U(n-k)}$$

given by the projection map $V_k(\mathbb{C}^n) \to \text{Gr}_k(\mathbb{C}^n)$.

5 The Categorical Viewpoint

5.1 The Category of Equivariant Spaces

Definition 5.1.1 (The Category of G-Spaces) Let G be a topological space. Define the category of G-spaces

$$_G\mathbf{Top}$$

to consist of the following data.

- The objects are the G-spaces
- The morphisms are the G-equivariant spaces
- Composition is given by the composition of functions.

There is an obvious forgetful functor $_G\mathbf{Top} \to \mathbf{Top}$. One of its adjoint should assign the space to a trivial G-action.

Definition 5.1.2 (The Trivial G-Space Functor) Let G be a topological group. Define the trivial G-space functor

$$\text{Triv} : \mathbf{Top} \to {_G\mathbf{Top}}$$

by the following.

- For each space X, define a group action on X by $g \cdot x = x$ for all $g \in G$ and $x \in X$.
- For each map $f : X \to Y$, $\text{Triv}(f) = f$ because f is trivially equivariant.

Note: inverses limits in $_G\mathbf{Top}$ always exists.

5.2 Induced and Restricted G-Spaces

Definition 5.2.1 (Induced G-Spaces) Let G be a topological group. Let $H \leq G$ be a subgroup. Let X be an H-space. Define the induced G-space of X to be the space

$$\text{Ind}_H^G X = G \times_H X = \frac{G \times X}{\sim}$$

where the relation is generated by $(g \cdot h, x) \sim (g, h \cdot x)$ for $g \in G$, $h \in H$ and $x \in X$.

Pushout?

Lemma 5.2.2 Let G be a topological group. Let $H \leq G$ be a subgroup. Let X be an H-space. Then the following are true.

- If $X = *$, then $G \times_H X \cong \frac{G}{H}$.
- If $h \cdot x = x$ for all $h \in H$ (the action of H is trivial), then $G \times_H X \cong \frac{G}{H} \times X$.

Definition 5.2.3 (Restricted G-Spaces) Let G be a topological group. Let $H \leq G$ be a subgroup. Let X be a G-space. Define the restriction

$$\text{Res}_H^G X$$

of X to H to be the space X considered as an H-space by the group action of G.

Proposition 5.2.4 Let G be a topological group. Let $H \leq G$ be a subgroup. Then there is an adjunction

$$\text{Ind}_H^G : {_H\mathbf{Top}} \rightleftarrows {_G\mathbf{Top}} : \text{Res}_H^G$$

This means that there is an isomorphism

$$\text{Hom}_{_G\mathbf{Top}}(\text{Ind}_H^G X, Y) \cong \text{Hom}_{_H\mathbf{Top}}(X, \text{Res}_H^G Y)$$

that is natural in X and Y.

5.3 Fixed Points and Orbit Spaces

Definition 5.3.1 (The Fixed Points Functor) Let G be a topological group. Define the fixed points functor

$$(-)^G : {}_G\mathbf{Top} \to \mathbf{Top}$$

by the following.

- For each G-space, X, $X^G = \{x \in X \mid g \cdot x = x \text{ for all } g \in G\}$ is the subset of fixed points of G equipped with the subspace topology.
- For each G-equivariant map $f : X :\to Y$, $(f)^G : X^G \to Y^G$ is the restriction of f to X^G.

Check: it is well defined.

Proposition 5.3.2 Let G be a topological group. There is an adjunction

$$\text{Triv} : \mathbf{Top} \rightleftarrows {}_G\mathbf{Top} : (-)^G$$

This means that there is an isomorphism

$$\text{Hom}_{{}_G\mathbf{Top}}(\text{Triv}X, Y) \cong \text{Hom}_{\mathbf{Top}}(X, Y^G)$$

that is natural in X and Y.

5.4 The Pointed Analogue

Definition 5.4.1 (Pointed G-Spaces)
Let G be a topological group. Let (X, x_0) be a pointed space. Suppose that X is a G-space. We say that (X, x_0) is a pointed G-space if x_0 is a fixed point of the action.

Definition 5.4.2 (Induced Pointed G-Spaces) Let G be a topological group. Let $H \leq G$ be a subgroup. Let (X, x_0) be a pointed H-space. Define the induced pointed G-space of (X, x_0) to be the space

$$\text{Ind}_H^G X = G_+ \wedge_H X = \frac{G_+ \wedge X}{\sim}$$

where the relation is generated by $(g \cdot h, x) \sim (g, h \cdot x)$ for all $g \in G$, $h \in H$ and $x \in X$.

Pushout?

Proposition 5.4.3 Let G be a topological group. Let $H \leq G$ be a subgroup. Then there is an adjunction

$$\text{Ind}_H^G : {}_H\mathbf{Top}_* \rightleftarrows {}_G\mathbf{Top}_* : \text{Res}_H^G$$

This means that there is an isomorphism

$$\text{Hom}_{{}_G\mathbf{Top}_*}(\text{Ind}_H^G X, Y) \cong \text{Hom}_{{}_H\mathbf{Top}_*}(X, \text{Res}_H^G Y)$$

that is natural in X and Y.